Diffusion-Driven Graph Reconstruction and Cross-View Contrastive Learning for Bundle Recommendation

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Abstract

Bundle recommendation integrates bundle-level feedback and item-level information to model users' preferences for sets of items. Recent research has integrated user-bundle (U-B) interactions, user-item behavior, and bundle composition into a common multi-view framework that combines graph-based modeling, representation fusion, and contrastive learning. Despite this progress, two challenges remain: (1) noisy U-B interactions can spread misleading signals through graph propagation; (2) sparse U-B feedback limits preference learning for users with few bundle interactions. To address these challenges, we propose ReCoBR, a framework that combines diffusion-driven graph reconstruction with cross-view contrastive learning. ReCoBR pretrains user and bundle embeddings, learns a user-conditioned latent diffusion model to refine aggregated historical bundle representations, and uses the refined preferences to re-score observed interactions and reconstruct the U-B graph. To adapt the denoising process to bundle recommendation, we introduce a bundle-set consistency loss that complements preference reconstruction by encouraging agreement between the refined preference and historical bundle embeddings. The reconstructed U-B view then anchors cross-view contrastive learning before representation fusion. User-side and bundle-side objectives connect U-B representations with their user-item and bundle-item counterparts, allowing behavioral and compositional information to support preference learning under sparse U-B feedback. Experiments on three public benchmarks show that ReCoBR outperforms the compared baselines.

1 Introduction

Bundle recommendation (BR) aims to match users with collections of related items, such as music playlists and fashion outfits (Chen et al. 2019a; Ma et al. 2022). For platforms, BR can create additional sales opportunities; for users, it can simplify the discovery and selection of collections suited to their needs (Ma et al. 2022, 2024). Learning the preferences underlying these recommendations involves modeling bundle-level choices through user-bundle (U-B) interactions and item-level behavior through user-item (U-I) interactions, with bundle-item (B-I) affiliations linking individual items to bundles. Effectively integrating information about user behavior and bundle composition is therefore key to accurately modeling users' preferences and providing personalized bundle recommendations.

Early approaches pursue this goal through joint learning at the bundle and item levels. DAM, for example, jointly models U-B and U-I interactions and uses attention to aggregate constituent-item representations into bundle representations (Chen et al. 2019a). Graph-based approaches further exploit the collaborative structure formed by these relations. BGCN uses graph propagation to model preferences at both the bundle and item levels, linking items to bundles through B-I affiliations (Chang et al. 2020). This graph formulation captures higher-order collaborative information while retaining two perspectives on users' bundle preferences.

Learning separate views also raises the question of how they can inform one another. CrossCBR addresses this question through cross-view contrastive learning, encouraging agreement between the representations of the same user or bundle across the bundle and item views (Ma et al. 2022). MultiCBR extends the formulation to three views by encoding U-B interactions, U-I interactions, and B-I affiliations separately (Ma et al. 2024). It combines these representations before prediction, allowing information from different views to interact in bundle scoring, and applies contrastive learning to the fused representation. DGMAE explores another form of cross-view cooperation by distilling first- and higher-order U-B knowledge into item-view representation learning (Ren et al. 2023b).

These advances demonstrate the benefits of coordinating complementary views, while also highlighting the central role of U-B interactions in bundle recommendation: U-B interactions define the bundle-view graph used for propagation, provide direct supervision for bundle ranking, and supply the bundle-level signal that is coordinated with auxiliary views in multi-view learning (Ma et al. 2022, 2024; Ren et al. 2023b). However, existing multi-view methods mainly focus on representation construction and cross-view coordination, typically treating the observed U-B feedback itself as a given rather than explicitly examining or refining the information it provides. At the same time, the reliability and sufficiency of observed U-B feedback remain concerns, as it can be noisy and sparse. Based on the above analysis, we identify the following two challenges.

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Challenge 1: Noise in user-bundle interactions. Implicit feedback may include interactions that do not faithfully reflect users' preferences (Wang et al. 2021; Jiang et al. 2026). Once incorporated as edges in the U-B graph, such interactions can introduce misleading signals that spread through neighborhood aggregation into the representations used for bundle ranking (Wu et al. 2021). At the same time, observed U-B interactions contain useful bundle-level evidence about each user's preferences. The challenge is therefore to limit the influence of potentially misleading connections during graph propagation without discarding useful personalized evidence.

Challenge 2: Sparse user-bundle feedback. Even reliable U-B feedback provides limited direct evidence when a user has interacted with only a few bundles. U-I interactions and B-I affiliations can supplement this evidence by revealing the user's item-level interests and the composition of bundles. However, interest in an individual item does not necessarily imply preference for the complete bundle (Ren et al. 2023b). The challenge is therefore to use this auxiliary information to supplement sparse U-B feedback while retaining users' bundle-level choices as the reference for preference learning.

To address these challenges, we propose **ReCoBR** (Reconstruction and Contrastive Learning for Bundle Recommendation), which combines diffusion-driven U-B graph reconstruction with cross-view contrastive learning. ReCoBR first pretrains user and bundle embeddings on the U-B graph with LightGCN (He et al. 2020) and aggregates each user's historical bundle embeddings into a preference representation. Conditioned on the pretrained user embedding, a latent diffusion model learns to recover this representation from Gaussian perturbations (Wang et al. 2023; Zhao et al. 2024), while a bundle-set consistency loss encourages the recovered preference to align with the user's historical bundle embeddings. The denoiser then iteratively refines the preference representation and re-scores observed interactions, retaining each user's highest-scoring connections for graph propagation; the original U-B interactions continue to provide ranking supervision. Using the reconstructed U-B graph alongside the U-I and B-I relations, ReCoBR learns view-specific user and bundle representations. Before fusion, U-B–U-I and U-B–B-I contrastive objectives jointly optimize the paired representations on both the user and bundle sides, with the reconstructed U-B view as their common reference. These objectives allow behavioral and compositional evidence to support bundle-level preference learning under sparse U-B feedback.

Our main contributions are as follows:

- We develop a diffusion-driven U-B graph reconstruction framework that integrates pretraining, user-conditioned historical preference refinement, and observed-interaction selection, supplemented by a bundle-set consistency objective.
- We introduce cross-view contrastive objectives that use the reconstructed U-B view as a common anchor for user-side and bundle-side learning from auxiliary views before fusion.
- Experiments on three public datasets demonstrate ReCoBR's recommendation improvements and examine the contributions of graph reconstruction and cross-view learning through ablations, interaction-selection comparisons, and user-group analysis.

2 Related Work

2.1 Bundle Recommendation

Bundle recommendation covers both predefined-bundle ranking and the generation of new item combinations (Sun et al. 2024). For predefined-bundle ranking, a shared goal is to combine direct bundle-level feedback with information about constituent items. Early approaches connect the two levels through attentive item aggregation and joint learning of U-B and U-I interactions (Chen et al. 2019a). Within graph-based modeling, neighborhood aggregation captures collaborative structure (He et al. 2020; Chang et al. 2020). Graph models also account for user motivations and bundling patterns: MIDGN disentangles user intents across global and local views (Zhao et al. 2022), while BundleGT models bundling strategies with hierarchical graph Transformers (Wei et al. 2023). Item-level evidence is also enriched through bundle–user–item correlations (Du et al. 2025) and relations among items (Li, Liu, and Wei 2026).

Constructing informative representations also raises the question of how information should be coordinated across views. CrossCBR aligns bundle and item views (Ma et al. 2022), whereas MultiCBR applies contrastive learning after three-view fusion (Ma et al. 2024). Distillation offers another form of cooperation by transferring first- and higher-order U-B knowledge to item-view learning (Ren et al. 2023b). ReCoBR further considers which observed U-B connections should be retained for propagation and cross-view preference learning.

2.2 Denoising and Diffusion for Recommendation

Denoising methods differ in how feedback reliability enters recommendation learning. Loss-based methods suppress potentially noisy feedback through truncation or reweighting (Wang et al. 2021), whereas cross-model agreement provides an alternative constraint on learned user preferences (Wang et al. 2022). For recommenders based on alignment and uniformity, CAFU constrains positive-pair alignment and filters unreliable pairs in the uniformity objective (Jiang et al. 2026).

Diffusion models learn a generative process that reverses controlled perturbations (Ho, Jain, and Abbeel 2020). In recommendation, key design choices concern the recovery target and conditioning context. DiffRec reconstructs interaction vectors, including through a latent-space variant (Wang et al. 2023), whereas DiffuRec recovers target-item embeddings conditioned on interaction sequences (Li, Sun, and Li 2023). For bundle recommendation, DCBR combines conditional U-B interaction denoising with disentangled contrastive learning (Li 2025), and RDiffBR applies residual diffusion to item-level bundle embeddings to accommodate composition changes (Zhang et al. 2026). ReCoBR uses user-conditioned denoising with bundle-set consistency to refine

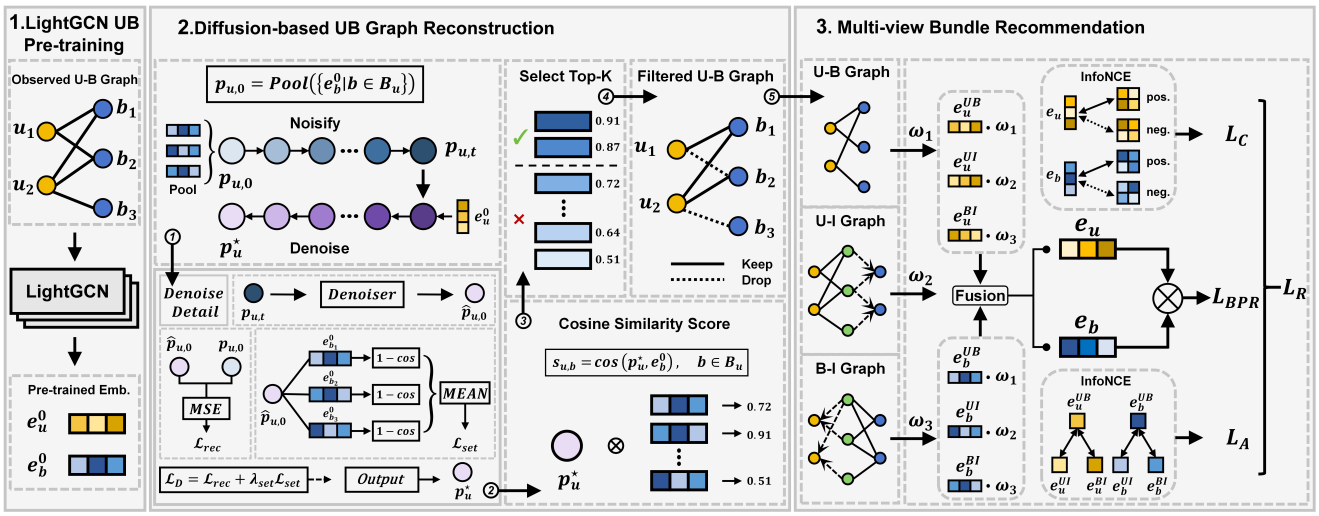


Figure 1: Overall framework of ReCoBR, comprising U-B pretraining, diffusion-driven U-B graph reconstruction with user-conditioned denoising and bundle-set consistency, and multi-view recommendation with pre-fusion cross-view contrastive learning.

historical bundle-preference representations and guide selection of observed U-B connections.

2.3 Contrastive Learning for Recommendation

Contrastive learning encourages agreement between related representations, making view construction central to the learning signal (Chen et al. 2020). Graph-based strategies construct views through random structural perturbations (Wu et al. 2021), or adapt them using learned interaction weights (Ren et al. 2023a) or paired graph-generative and graph-denoising models (Jiang, Huang, and Xia 2023). At the objective level, BIGCF couples augmentation-free contrastive regularization in interaction and intent spaces with Gaussian graph generation and bilateral intent modeling (Zhang, Sang, and Zhang 2024).

Auxiliary relations provide another source of contrastive supervision. In knowledge-aware recommendation, knowledge-graph consistency guides interaction augmentation (Yang et al. 2022), while multi-level objectives connect collaborative, semantic, and structural views (Zou et al. 2022). These approaches emphasize both how views are formed and which relationships are enforced between their representations. ReCoBR applies cross-view objectives before fusion, using the reconstructed U-B view as the common reference for learning from behavioral and compositional views.

3 Methodology

3.1 Task Definition and Framework Overview

Let $\mathcal{U}$, $\mathcal{B}$, and $\mathcal{I}$ denote the sets of users, predefined bundles, and items, respectively. The available training data comprise three binary matrices: user-bundle interactions $\mathbf{X} \in \{0,1\}^{|\mathcal{U}|\times|\mathcal{B}|}$, user-item interactions $\mathbf{Y} \in \{0,1\}^{|\mathcal{U}|\times|\mathcal{I}|}$, and bundle-item affiliations $\mathbf{Z} \in \{0,1\}^{|\mathcal{B}|\times|\mathcal{I}|}$. In $\mathbf{X}$ and $\mathbf{Y}$, an entry equal to one denotes an observed interaction; in $\mathbf{Z}$, it denotes bundle membership. Given these relations, our task is to estimate a preference score $\hat{y}_{ub}$ for each unobserved user-bundle pair and rank candidate bundles for each user.

Figure 1 outlines ReCoBR. To address noise in U-B interactions, we first pretrain user and bundle representations, learn a user-conditioned denoiser over historical bundle representations, and use the refined preferences to reconstruct the U-B graph. The denoising objective incorporates a bundle-set consistency loss (SetLoss) to retain bundle-preference information during recovery. To address sparse U-B feedback, we then use the reconstructed U-B view as a common anchor for cross-view contrastive learning before representation fusion. The fused user and bundle representations produce the final recommendation scores.

We distinguish the observed matrix $\mathbf{X}$ from the reconstructed matrix $\widetilde{\mathbf{X}}$ throughout this section. The former defines users' training histories and ranking examples; the latter determines the U-B connections used for recommendation graph propagation.

3.2 Diffusion-Driven U-B Graph Denoising and Reconstruction

The purpose of this module is to evaluate an observed interaction in the context of the user's historical bundle preferences. We construct this context in a pretrained representation space, refine it through conditional denoising, and translate the resulting preference vector into a selection of graph connections.

Pretraining and Historical Preference Construction We pretrain LightGCN (He et al. 2020) on the observed U-B graph using Bayesian Personalized Ranking (BPR) (Rendle et al. 2009). Let $\mathbf{h}_u^{(0)}, \mathbf{h}_b^{(0)} \in \mathbb{R}^d$ be its trainable initial embeddings, and let $\mathcal{N}_u = \{b \in \mathcal{B} : x_{ub} = 1\}$ and $\mathcal{N}_b = \{u \in \mathcal{U} : x_{ub} = 1\}$ denote the observed neighbors

of user u and bundle b. LightGCN propagation and layer aggregation are given by

$$\begin{aligned}\mathbf{h}_u^{(\ell+1)} &= \sum_{b\in\mathcal{N}_u} \frac{\mathbf{h}_b^{(\ell)}}{\sqrt{|\mathcal{N}_u||\mathcal{N}_b|}},\\ \mathbf{h}_b^{(\ell+1)} &= \sum_{u\in\mathcal{N}_b} \frac{\mathbf{h}_u^{(\ell)}}{\sqrt{|\mathcal{N}_b||\mathcal{N}_u|}},\\ \mathbf{e}_u^0 &= \sum_{\ell=0}^{L_p} \eta_\ell \mathbf{h}_u^{(\ell)}, \qquad \mathbf{e}_b^0 = \sum_{\ell=0}^{L_p} \eta_\ell \mathbf{h}_b^{(\ell)}.\end{aligned} \tag{1}$$

Here, L_p is the number of pretraining propagation layers, and the fixed aggregation coefficients satisfy $\eta_\ell \geq 0$ and $\sum_{\ell=0}^{L_p} \eta_\ell = 1$. The resulting embeddings $\mathbf{e}_u^0$ and $\mathbf{e}_b^0$ remain fixed during graph reconstruction. They are subsequently used to construct historical preferences, condition the denoiser, and score observed interactions.

For each active user u, we summarize the representations of the historically interacted bundles by mean pooling:

$$\mathbf{p}_{u,0} = \frac{1}{|\mathcal{N}_u|} \sum_{b\in\mathcal{N}_u} \mathbf{e}_b^0. \tag{2}$$

The vector $\mathbf{p}_{u,0}$ represents the user's observed bundle-preference context. Mean pooling accommodates histories of different lengths while keeping the denoising input d-dimensional. We construct this vector for users in $\mathcal{U}_+ = \{u \in \mathcal{U} : |\mathcal{N}_u| > 0\}$. The subscript 0 denotes the state before artificial noise is added; it does not assume that the observed history is free of interaction noise.

User-Conditioned Diffusion Denoising We train the denoiser to recover historical preference representations under different levels of perturbation. Following the Gaussian diffusion formulation (Ho, Jain, and Abbeel 2020; Wang et al. 2023), let T be the number of diffusion steps and $\beta_t \in (0,1)$ the noise variance at step t. We use a linear variance schedule and define $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. A perturbed preference vector can be sampled directly as

$$\mathbf{p}_{u,t} = \sqrt{\bar{\alpha}_t}\,\mathbf{p}_{u,0} + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d). \tag{3}$$

Here, $\boldsymbol{\epsilon} \in \mathbb{R}^d$ is standard Gaussian noise, $\mathbf{I}_d$ is the d-dimensional identity matrix, and $\bar{\alpha}_t$ controls how much historical information remains at noise level t.

To make recovery personalized, we condition the denoiser on the pretrained user representation $\mathbf{e}_u^0$. We implement the denoiser f_θ as a multilayer perceptron (MLP), with parameters θ, that predicts the unperturbed preference vector:

$$\begin{aligned}\widehat{\mathbf{p}}_{u,0} &= f_\theta(\mathbf{p}_{u,t}, \mathbf{e}_u^0, t),\\ f_\theta(\mathbf{p}, \mathbf{c}, t) &= \mathrm{MLP}_\theta\big([\mathbf{p}; \mathbf{c}; \mathbf{p}\odot\mathbf{c}; |\mathbf{p}-\mathbf{c}|; \boldsymbol{\psi}(t)]\big).\end{aligned} \tag{4}$$

In Eq. (4), $[\cdot;\cdot]$ denotes concatenation, $\odot$ denotes element-wise multiplication, and $|\mathbf{p}-\mathbf{c}|$ is the element-wise absolute deviation. The vectors $\mathbf{p}$ and $\mathbf{c}$ preserve the current preference estimate and the user condition, while $\mathbf{p}\odot\mathbf{c}$ and $|\mathbf{p}-\mathbf{c}|$ expose their coordinate-wise agreement and discrepancy. The sinusoidal embedding $\boldsymbol{\psi}(t) \in \mathbb{R}^{d_t}$ specifies the perturbation level. Together with the timestep embedding, these features allow the MLP to adapt preference recovery to both the user and the perturbation level. Consequently, the MLP receives a $(4d+d_t)$-dimensional input and predicts a d-dimensional preference vector.

During training, we uniformly sample $u \in \mathcal{U}_+$ and $t \in \{1,\dots,T\}$ and draw Gaussian noise as in Eq. (3). The reconstruction objective is

$$\mathcal{L}_{\mathrm{rec}} = \mathbb{E}_{u,t,\boldsymbol{\epsilon}}\left[\frac{1}{d}\left\|\widehat{\mathbf{p}}_{u,0} - \mathbf{p}_{u,0}\right\|_2^2\right]. \tag{5}$$

This objective learns a shared recovery function from perturbed historical preferences. The original representation supplies the reconstruction target, while the pretrained user embedding provides personalized context.

Bundle-Set Consistency in Denoising Reconstructing the pooled vector provides a basic denoising objective, but subsequent interaction selection depends on the recovered vector's compatibility with individual historical bundles. We therefore introduce SetLoss to connect preference recovery with the bundle representations that will be used for graph reconstruction. Figure 1 illustrates how SetLoss complements the reconstruction objective in denoiser training.

Let $\mathrm{sim}(\mathbf{a},\mathbf{b}) = \mathbf{a}^\top\mathbf{b}/(\|\mathbf{a}\|_2\|\mathbf{b}\|_2)$ denote cosine similarity, with a small numerical safeguard for zero norms. SetLoss encourages average directional agreement between the recovered preference and the user's historical bundle representations:

$$\mathcal{L}_{\mathrm{set}} = \mathbb{E}_{u,t,\boldsymbol{\epsilon}}\left[\frac{1}{|\mathcal{N}_u|}\sum_{b\in\mathcal{N}_u}\left(1-\mathrm{sim}\left(\widehat{\mathbf{p}}_{u,0}, \mathbf{e}_b^0\right)\right)\right]. \tag{6}$$

Equation (5) reconstructs the pooled preference vector, whereas Eq. (6) aligns the recovered preference with its constituent historical bundles under the cosine-similarity geometry later used for interaction scoring. Averaging over the history prevents users with more interactions from dominating the objective.

We optimize the denoiser with

$$\mathcal{L}_{\mathrm{D}} = \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{set}}\mathcal{L}_{\mathrm{set}}, \tag{7}$$

where $\lambda_{\mathrm{set}} \geq 0$ controls the contribution of SetLoss.

Preference Refinement and Graph Reconstruction After learning the denoiser, we use it to refine the historical preference vector for interaction scoring. This stage employs a deterministic refinement procedure initialized with $\mathbf{p}_{u,0}$. Specifically,

$$\begin{aligned}\mathbf{r}_{u,T} &= \mathbf{p}_{u,0},\\ \mathbf{r}_{u,t-1} &= f_\theta(\mathbf{r}_{u,t}, \mathbf{e}_u^0, t), \qquad t = T, T-1, \dots, 1,\\ \mathbf{p}_u^\star &= \mathbf{r}_{u,0}.\end{aligned} \tag{8}$$

The states $\mathbf{r}_{u,t}$ are refinement iterates, distinct from the forward-corrupted training states $\mathbf{p}_{u,t}$. Each iteration feeds the current estimate back into the denoiser with the next timestep condition. Thus, graph reconstruction applies the learned recovery network to an observed preference context, without sampling a Gaussian starting state.

We measure how closely each observed bundle matches the refined preference:

$$s_{ub} = \mathrm{sim}(\mathbf{p}_u^\star, \mathbf{e}_b^0), \qquad b \in \mathcal{N}_u. \tag{9}$$

For each user, we retain the highest-scoring $\kappa_u = \min(\kappa, |\mathcal{N}_u|)$ bundles, where the positive integer κ controls the maximum retained degree:

$$\begin{aligned} \widetilde{\mathcal{N}}_u &= \underset{b \in \mathcal{N}_u}{\mathrm{TopK}}\left(s_{ub}, \kappa_u\right), \\ \widetilde{x}_{ub} &= \mathbf{1}\left[b \in \widetilde{\mathcal{N}}_u\right]. \end{aligned} \tag{10}$$

Here, $\mathrm{TopK}(s_{ub}, \kappa_u)$ returns the set of κ_u bundles in $\mathcal{N}_u$ with the largest scores, and $\mathbf{1}[\cdot]$ is the indicator function. Users with at most κ observed bundles retain their full histories; users outside $\mathcal{U}_+$ have no retained U-B edges. Selection is restricted to observed interactions, so graph reconstruction neither creates new edges nor changes the ranking labels defined by $\mathbf{X}$. The resulting binary matrix $\widetilde{\mathbf{X}}$ is the reconstructed U-B graph used for downstream propagation.

3.3 Pre-Fusion Cross-View Contrastive Learning

With the U-B graph reconstructed, we next learn representations from this graph together with user-item behavior and bundle composition. The auxiliary relations provide complementary evidence when a user's bundle feedback is limited. We first construct user and bundle representations in each view, and then connect them through U-B-anchored contrastive learning before fusion.

View-Specific Representations Following the three-view backbone of MultiCBR (Ma et al. 2024), we construct the U-B, U-I, and B-I propagation graphs from $\widetilde{\mathbf{X}}$, $\mathbf{Y}$, and $\mathbf{Z}$, respectively. Each encoder applies L layers of symmetrically normalized LightGCN-style propagation (He et al. 2020). Fixed view-specific coefficients combine the initial embeddings with the ℓ_2-normalized layer outputs. During training, the inherited backbone adds small representation perturbations to the propagation outputs and auxiliary item pooling; these perturbations are disabled during evaluation. The encoders share trainable initial embeddings for the same entity, which are separate from the fixed pretrained embeddings in Section 3.2.

U-B view. Propagation on the reconstructed U-B graph directly produces user representations $\mathbf{e}_u^{\mathrm{UB}}$ and bundle representations $\mathbf{e}_b^{\mathrm{UB}}$, capturing collaborative information through the retained interactions.

U-I view. Propagation on the U-I graph produces user representations $\mathbf{e}_u^{\mathrm{UI}}$ and item representations $\mathbf{e}_i^{\mathrm{UI}}$. To obtain a bundle representation $\mathbf{e}_b^{\mathrm{UI}}$ in this view, we average the representations of its constituent items according to $\mathbf{Z}$.

B-I view. Propagation on the B-I graph produces bundle representations $\mathbf{e}_b^{\mathrm{BI}}$ and item representations $\mathbf{e}_i^{\mathrm{BI}}$. To obtain a user representation $\mathbf{e}_u^{\mathrm{BI}}$ in this view, we average the representations of the items that the user has interacted with according to $\mathbf{Y}$.

The two item-pooling operations are

$$\mathbf{e}_b^{\mathrm{UI}} = \frac{1}{|\mathcal{I}_b|} \sum_{i \in \mathcal{I}_b} \mathbf{e}_i^{\mathrm{UI}}, \qquad \mathbf{e}_u^{\mathrm{BI}} = \frac{1}{|\mathcal{I}_u|} \sum_{i \in \mathcal{I}_u} \mathbf{e}_i^{\mathrm{BI}}, \tag{11}$$

where $\mathcal{I}_b = \{i : z_{bi} = 1\}$ contains the items in bundle b, and $\mathcal{I}_u = \{i : y_{ui} = 1\}$ contains the items interacted with by user u. An empty neighborhood contributes a zero vector.

This step yields three representations for each user, $\{\mathbf{e}_u^{\mathrm{UB}}, \mathbf{e}_u^{\mathrm{UI}}, \mathbf{e}_u^{\mathrm{BI}}\}$, and each bundle, $\{\mathbf{e}_b^{\mathrm{UB}}, \mathbf{e}_b^{\mathrm{UI}}, \mathbf{e}_b^{\mathrm{BI}}\}$, all in $\mathbb{R}^d$. These unfused representations are used to construct the cross-view objective below and are later fused for prediction.

Cross-View Objectives Anchored in U-B Using these view-specific representations, we pair each user's U-B representation with its U-I and B-I counterparts, and do the same for each bundle. The reconstructed U-B view thus serves as the common reference. Each same-entity pair is a positive pair, while other entities of the same type provide in-batch negatives. For every recommendation minibatch, we remove duplicate user IDs and duplicate positive-bundle IDs to form the entity sets $\mathcal{S}_U$ and $\mathcal{S}_B$, ensuring that an entity is not repeated as its own in-batch negative.

For two views v and w and an entity set $\mathcal{S}$, we define the directional InfoNCE objective (van den Oord, Li, and Vinyals 2018) as

$$\ell_\tau^{v,w}(\mathcal{S}) = -\frac{1}{|\mathcal{S}|} \sum_{q \in \mathcal{S}} \log \frac{\exp\big(\mathrm{sim}(\mathbf{e}_q^v, \mathbf{e}_q^w)/\tau\big)}{\sum_{q' \in \mathcal{S}} \exp\Big(\mathrm{sim}(\mathbf{e}_q^v, \mathbf{e}_{q'}^w)/\tau\Big)}. \tag{12}$$

The temperature $\tau > 0$ controls sensitivity to similarity differences. For each anchor $\mathbf{e}_q^v$, its counterpart $\mathbf{e}_q^w$ is the positive candidate, and the representations of all entities in $\mathcal{S}$, including the positive, form the denominator. The pre-fusion objective is

$$\begin{aligned} \mathcal{L}_{\mathrm{A}} = \ell_\tau^{\mathrm{UB,UI}}(\mathcal{S}_U) + \ell_\tau^{\mathrm{UB,BI}}(\mathcal{S}_U) \\ + \ell_\tau^{\mathrm{UB,UI}}(\mathcal{S}_B) + \ell_\tau^{\mathrm{UB,BI}}(\mathcal{S}_B). \end{aligned} \tag{13}$$

Equation (13) encourages agreement for the same user or bundle across U-B–U-I and U-B–B-I pairs while preserving discrimination between entities. A user-side or bundle-side term is omitted when the corresponding entity set contains fewer than two members and therefore provides no in-batch negatives. The U-B view is the common anchor in constructing the objective, and gradients update both representations in each pair. These constraints inject behavioral and compositional evidence into bundle-level preference learning when U-B histories are sparse.

3.4 Prediction and Model Optimization

View fusion and prediction. Following MultiCBR (Ma et al. 2024), we adopt its fixed view coefficients to combine the three view-specific representations into a single user representation and a single bundle representation. Their inner product gives the preference score:

$$\begin{aligned} \mathbf{e}_u &= \omega_1 \mathbf{e}_u^{\mathrm{UB}} + \omega_2 \mathbf{e}_u^{\mathrm{UI}} + \omega_3 \mathbf{e}_u^{\mathrm{BI}}, \\ \mathbf{e}_b &= \omega_1 \mathbf{e}_b^{\mathrm{UB}} + \omega_2 \mathbf{e}_b^{\mathrm{UI}} + \omega_3 \mathbf{e}_b^{\mathrm{BI}}, \\ \hat{y}_{ub} &= \mathbf{e}_u^\top \mathbf{e}_b. \end{aligned} \tag{14}$$

Here, ω_1, ω_2, and ω_3 are fixed coefficients for the U-B, U-I, and B-I views, respectively. The same coefficients are used

for users and bundles; they are nonnegative and satisfy $\omega_1 + \omega_2 + \omega_3 = 1$.

Recommendation objectives. Let $\mathcal{Q}$ be a minibatch of m ranking triplets, indexed as (u_j, b_j^+, b_j^-) for $j = 1, \ldots, m$. Here, $x_{u_j b_j^+} = 1$ and $b_j^- \in \mathcal{B} \setminus \mathcal{N}_{u_j}$. Following MultiCBR (Ma et al. 2024), the fused-space self-contrastive regularizer is

$$\mathcal{L}_{\mathrm{C}} = -\frac{1}{2m} \sum_{j=1}^{m} \left[\log \frac{\exp\big(\mathrm{sim}(\mathbf{e}_{u_j}, \mathbf{e}_{u_j})/\tau\big)}{\sum_{k=1}^{m} \exp\big(\mathrm{sim}(\mathbf{e}_{u_j}, \mathbf{e}_{u_k})/\tau\big)} + \log \frac{\exp\Big(\mathrm{sim}(\mathbf{e}_{b_j^+}, \mathbf{e}_{b_j^+})/\tau\Big)}{\sum_{k=1}^{m} \exp\Big(\mathrm{sim}(\mathbf{e}_{b_j^+}, \mathbf{e}_{b_k^+})/\tau\Big)} \right]. \tag{15}$$

For each occurrence in the minibatch, the numerator pairs its fused representation with itself, while all minibatch occurrences of the same entity type appear in the denominator. This inherited objective regularizes separation in the fused representation space, whereas $\mathcal{L}_{\mathrm{A}}$ explicitly connects different relation views before fusion. We optimize BPR (Rendle et al. 2009) together with the two regularization objectives:

$$\mathcal{L}_{\mathrm{BPR}} = -\frac{1}{m} \sum_{j=1}^{m} \log \sigma\Big(\hat{y}_{u_j b_j^+} - \hat{y}_{u_j b_j^-}\Big), \qquad \mathcal{L}_{\mathrm{R}} = \mathcal{L}_{\mathrm{BPR}} + \lambda_a \mathcal{L}_{\mathrm{A}} + \lambda_c \mathcal{L}_{\mathrm{C}} + \lambda_r \|\Theta\|_2^2. \tag{16}$$

Here, σ is the sigmoid function, Θ comprises the trainable initial user, bundle, and item embeddings of the recommendation backbone, and $\lambda_a, \lambda_c, \lambda_r \geq 0$ balance the loss terms. Ranking examples continue to come from $\mathbf{X}$, even when their edges are not selected for $\widetilde{\mathbf{X}}$. Graph reconstruction therefore filters propagation connections while preserving the observed feedback used for ranking.

Training. After offline U-B pretraining, each training epoch contains three stages. First, we update the denoiser f_θ for one pass over $\mathcal{U}_+$ using $\mathcal{L}_{\mathrm{D}}$. Second, the updated denoiser refines every active user's historical preference, scores the observed U-B interactions, and refreshes $\widetilde{\mathbf{X}}$ through the Top-K selection in Eq. (10). Third, we hold $\widetilde{\mathbf{X}}$ fixed and update the recommendation parameters Θ on ranking minibatches using $\mathcal{L}_{\mathrm{R}}$. The denoiser and recommendation backbone use separate optimizers. Because Top-K edge selection is discrete, gradients from $\mathcal{L}_{\mathrm{R}}$ do not propagate through $\widetilde{\mathbf{X}}$ to f_θ.

Inference. Using the reconstructed graph associated with the selected model, we compute and cache the three view-specific representations and their fused user and bundle embeddings. For each user, we rank candidate bundles by Eq. (14) after excluding bundles observed in the training data. Graph reconstruction is performed before scoring; the denoiser is not run separately for each candidate bundle.

3.5 Computational Complexity

Let $n_+ = |\mathcal{U}_+|$ and $h_{\max} = \max_{u \in \mathcal{U}_+} |\mathcal{N}_u|$. Let $M = |\mathcal{E}_{\mathrm{UB}}|$ be the number of observed U-B edges, and let $E = |\widetilde{\mathcal{E}}_{\mathrm{UB}}| + |\mathcal{E}_{\mathrm{UI}}| + |\mathcal{E}_{\mathrm{BI}}|$ be the total number of edges used for graph propagation. We use Q and m to denote the number and size of recommendation minibatches, respectively. With a fixed-depth denoiser whose hidden widths scale with d, one evaluation costs $C_f = O(d^2)$. The time complexity of one training epoch is

$$O\begin{pmatrix} (T+1)n_+ C_f \\ + n_+ h_{\max}[d + \log(1+\kappa)] \\ + Q\big(LEd + m^2 d\big) \end{pmatrix}. \tag{17}$$

The three terms correspond to denoiser training and refinement, history processing and bounded-heap Top-K graph reconstruction, and recommendation learning, respectively. Offline LightGCN pretraining requires $O(L_p M d)$ per propagation pass. Since $|\widetilde{\mathcal{E}}_{\mathrm{UB}}| \leq M$, reconstruction does not densify the U-B graph.

For space complexity, let $V = |\mathcal{U}| + |\mathcal{B}| + |\mathcal{I}|$, let b_d be the maximum denoising or reconstruction batch size, and let P_θ be the denoiser parameter count. The peak training space is

$$O\begin{pmatrix} (L+1)Vd + P_\theta + (|\mathcal{U}| + |\mathcal{B}|)d \\ + E + M + m^2 + b_d h_{\max} d \end{pmatrix}. \tag{18}$$

At inference, constructing the cached graph representations costs $O(LEd)$. Scoring one user-bundle pair costs $O(d)$, and ranking all candidate bundles for one user costs $O(|\mathcal{B}|d)$.

4 Experiments

In this section, we evaluate ReCoBR against state-of-the-art bundle recommendation methods on three public datasets.

4.1 Experimental Settings

Datasets. To evaluate ReCoBR, we conduct experiments on three public datasets: **Youshu** (Chen et al. 2019a), **iFashion** (Chen et al. 2019b), and **NetEase** (Cao et al. 2017). Table 1 summarizes their statistics. For a fair comparison, we use the same data splits and full-ranking evaluation protocol as MultiCBR (Ma et al. 2024). Performance is measured using Recall@K and NDCG@K (Wang et al. 2019) for $K \in \{20, 40\}$ under full-ranking evaluation, with training bundles excluded from the candidates.

Baselines. We compare ReCoBR with eleven representative recommendation methods. **LightGCN** (He et al. 2020) learns collaborative representations through neighborhood aggregation. **SGL** (Wu et al. 2021) applies contrastive learning to augmented graphs. **XSimGCL** (Yu et al. 2024)

Dataset	#Users	#Items	#Bundles	#U–I	#U–B	#Avg. I/B
Youshu	8,039	32,770	4,771	138,515	51,377	37.03
NetEase	18,528	123,628	22,864	1,128,065	302,303	77.80
iFashion	53,897	42,563	27,694	2,290,645	1,679,708	3.86

Table 1: Dataset statistics. U–I and U–B count user–item and user–bundle interactions; Avg. I/B is the average number of items per bundle.

Method	Youshu				iFashion				NetEase			
	R@20	N@20	R@40	N@40	R@20	N@20	R@40	N@40	R@20	N@20	R@40	N@40
LightGCN	0.2286	0.1344	0.3190	0.1592	0.0837	0.0612	0.1284	0.0770	0.0496	0.0254	0.0795	0.0334
SGL	0.2622	0.1551	0.3630	0.1825	0.1000	0.0739	0.1498	0.0915	0.0716	0.0386	0.1104	0.0488
XSimGCL	0.2644	0.1560	0.3569	0.1814	0.1003	0.0744	0.1501	0.0920	0.0710	0.0376	0.1079	0.0475
LightGCL	0.2710	0.1556	0.3693	0.1827	0.0894	0.0653	0.1362	0.0818	0.0705	0.0371	0.1122	0.0481
BGCN	0.2436	0.1329	0.3379	0.1589	0.0802	0.0582	0.1239	0.0736	0.0626	0.0328	0.0988	0.0424
MIDGN	0.2682	0.1527	0.3712	0.1808	0.0856	0.0473	0.1299	0.0593	0.0678	0.0343	0.1085	0.0451
CrossCBR	0.2823	0.1670	0.3787	0.1939	0.1132	0.0872	0.1623	0.1045	0.0844	0.0458	0.1255	0.0567
BundleGT	0.2905	0.1737	0.3909	0.2011	0.1263	0.0969	0.1819	0.1164	0.0903	0.0478	0.1389	0.0606
MultiCBR	0.2842	0.1693	0.3946	0.1994	0.1500	0.1203	0.2033	0.1390	0.0907	0.0494	0.1355	0.0612
EBRec	0.2858	0.1701	0.3871	0.1979	0.1329	0.1047	0.1873	0.1238	0.0899	0.0487	0.1359	0.0604
EpicCBR	0.2852	0.1691	0.3949	0.1985	0.1474	0.1189	0.2000	0.1374	0.0913	0.0502	0.1345	0.0615
ReCoBR	**0.2920**	**0.1759**	**0.3997**	**0.2023**	**0.1568**	**0.1315**	**0.2067**	**0.1491**	**0.0955**	**0.0511**	**0.1390**	**0.0627**
Improv.	**0.52%**	**1.27%**	**1.22%**	**0.60%**	**4.53%**	**9.31%**	**1.67%**	**7.27%**	**4.60%**	**1.79%**	**0.07%**	**1.95%**

Table 2: Overall performance at Top-20 and Top-40 on three datasets. R and N denote Recall and NDCG. Bold and underlined entries mark the best and second-best results, respectively. Improv. is ReCoBR's relative improvement over the strongest baseline for each metric.

Variant	Youshu		NetEase		iFashion	
	R@20	N@20	R@20	N@20	R@20	N@20
ReCoBR	**0.2920**	**0.1759**	**0.0955**	**0.0511**	**0.1568**	**0.1315**
w/o Graph Recon.	0.2919	0.1734	0.0948	0.0509	0.1500	0.1245
w/o SetLoss	0.2913	0.1749	0.0951	0.0506	0.1559	0.1301
w/o All CL	0.2326	0.1307	0.0512	0.0260	0.0400	0.0281
w/o Pre-fusion CL	0.2901	0.1692	0.0906	0.0487	0.1558	0.1304
w/o Post-fusion CL	0.2595	0.1502	0.0682	0.0350	0.0461	0.0350

Table 3: Component ablations on three datasets, evaluated by Recall@20 (R@20) and NDCG@20 (N@20). Bold entries identify the complete ReCoBR model.

uses noise-based embedding augmentation, while **LightGCL** (Cai et al. 2023) constructs contrastive views using singular value decomposition. These four methods treat bundles as items and use only U-B interactions. **BGCN** (Chang et al. 2020) models bundle- and item-level preferences through graph convolution, while **MIDGN** (Zhao et al. 2022) disentangles global and local intents. **CrossCBR** (Ma et al. 2022) aligns bundle and item views through contrastive learning. **BundleGT** (Wei et al. 2023) models bundling strategies with hierarchical graph Transformers. **MultiCBR** (Ma et al. 2024) fuses three relation views before contrastive learning. **EBRec** (Du et al. 2025) enhances item-level bundle representations using bundle–user–item correlations. **EpicCBR** (Li, Liu, and Wei 2026) combines item-relation enhancement with contrastive learning.

Results for LightGCN, SGL, XSimGCL, LightGCL, BGCN, MIDGN, CrossCBR, and MultiCBR are taken from MultiCBR (Ma et al. 2024); BundleGT and EBRec results are from their respective papers (Wei et al. 2023; Du et al. 2025). We reproduce EpicCBR using its open-source implementation.

Parameter Settings. We implement ReCoBR in PyTorch on a GeForce RTX 3090 GPU. We use 64-dimensional embeddings with Xavier (Glorot and Bengio 2010) initialization and two propagation layers. Recommendation training uses Adam (Kingma and Ba 2015) with learning rate 10^{-3}, batch size 2048, and one negative bundle per positive interaction; $\tau = 0.20$ and $\lambda_r = 10^{-5}$. The denoiser is a two-layer MLP with hidden size 128, timestep embedding size 16, and dropout 0.1, trained with batch size 1024, learning rate 10^{-3}, and weight decay 10^{-5}. The minimum noise variance is fixed at $\beta_1 = 10^{-4}$.

For sensitivity analysis, we vary β_T in $\{0.005, 0.01, 0.02, 0.04, 0.06, 0.08\}$, κ in $\{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$, λ_a in $\{0, 0.001, 0.003, 0.005, 0.007, 0.009\}$, λ_{set} in $\{0, 0.005, 0.01, 0.02, 0.04, 0.05, 0.10, 0.20\}$, and λ_c in $\{0, 0.01, 0.02, 0.04, 0.05, 0.10, 0.20, 0.40\}$. ReCoBR's overall results are averaged over three independent runs.

4.2 Overall Performance

Table 2 compares ReCoBR with eleven baselines. We draw the following observations:

- ReCoBR achieves the best results across all datasets and metrics. Its Recall@20 improves over the strongest baseline by 0.52%, 4.53%, and 4.60% on Youshu, iFashion, and NetEase, respectively. The gains are most pronounced on iFashion, particularly in NDCG. This may be because its richer U-B histories provide more context for preference-guided graph reconstruction and neighbor selection, helping distill clearer and more discriminative user preference signals.
- SGL, XSimGCL, and LightGCL outperform LightGCN across all datasets and metrics. On iFashion, they also exceed BGCN and MIDGN despite using only U-B interactions. These results show that incorporating item information alone does not guarantee better recommendation.

They highlight the importance of effective U-B preference learning, which ReCoBR addresses through preference-guided graph reconstruction.

- ReCoBR also outperforms CrossCBR, BundleGT, MultiCBR, EBRec, and EpicCBR, which build on the original U-B relations to model bundle preferences and integrate information across views. ReCoBR further considers which U-B connections should guide learning across views, bringing interaction refinement and auxiliary-information integration into one framework. ReCoBR's consistent gains across all three datasets support the value of diffusion-driven graph reconstruction in uncovering informative U-B preference signals for cross-view learning.

4.3 Ablation Study

Table 3 compares ReCoBR with five variants: **w/o Graph Recon.** restores the original U-B graph; **w/o SetLoss** removes the bundle-set consistency loss; **w/o Pre-fusion CL** and **w/o Post-fusion CL** remove their respective contrastive objectives; and **w/o All CL** removes both. Graph reconstruction improves ranking performance, particularly on iFashion, supporting preference-guided selection of propagation connections. SetLoss yields further gains, reinforcing the value of consistency with historical bundle preferences. Removing pre-fusion learning also degrades performance, highlighting the value of connecting U-B preferences with auxiliary views before fusion. The larger losses without post-fusion or all contrastive learning emphasize the importance of representation regularization. Overall, ReCoBR outperforms all variants, demonstrating the effectiveness of jointly refining U-B preference signals and integrating auxiliary information for bundle recommendation.

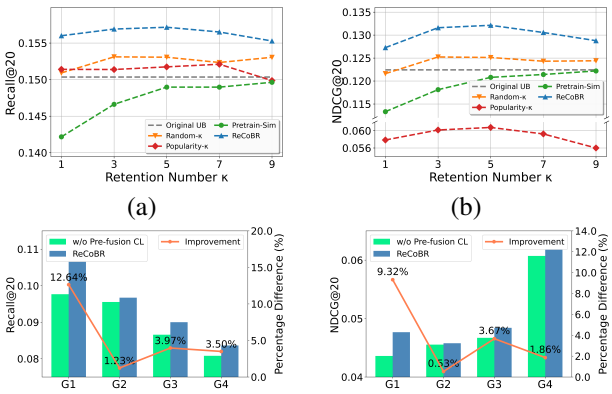


Figure 2: In-depth analysis of ReCoBR using Recall@20 and NDCG@20. (a)–(b) Interaction selection on iFashion ($K = \kappa$). (c)–(d) ReCoBR versus w/o Pre-fusion CL on NetEase, both retaining graph reconstruction. Users are sorted by increasing training U-B interaction counts and divided into four equally sized groups (G1–G4). Lines show relative improvements over w/o Pre-fusion CL.

4.4 In-depth Analysis

Interaction Selection Strategies. On iFashion, we compare ReCoBR with the original graph (**Original UB**) and three filtering strategies: random selection (**Random-K**), bundle popularity (**Popularity-K**), and pretrained user-bundle similarity (**Pretrain-Sim**). All filtering strategies select from users' training histories under the same per-user retention limit κ. As shown in Figure 2(a)–(b), ReCoBR consistently performs best. Random-K can improve over Original UB but remains inferior to ReCoBR, indicating that selecting appropriate connections matters beyond reducing their number. Popularity-K's weak NDCG suggests that global popularity is insufficient for personalized ranking. Pretrain-Sim even underperforms Random-K, suggesting that pretrained user-bundle similarity does not necessarily reflect a connection's value for subsequent propagation. ReCoBR instead evaluates each observed bundle against a refined historical preference representation, providing an explicit bundle-preference reference for neighbor selection.

Performance on Sparse Users. NetEase users are sorted by increasing training U-B interaction counts and divided into four equally sized groups (G1–G4). We compare ReCoBR with **w/o Pre-fusion CL**, retaining graph reconstruction in both. Figure 2(c)–(d) shows improvements across all four groups, with the largest relative gains in the sparsest group G1: 12.64% in Recall@20 and 9.32% in NDCG@20.

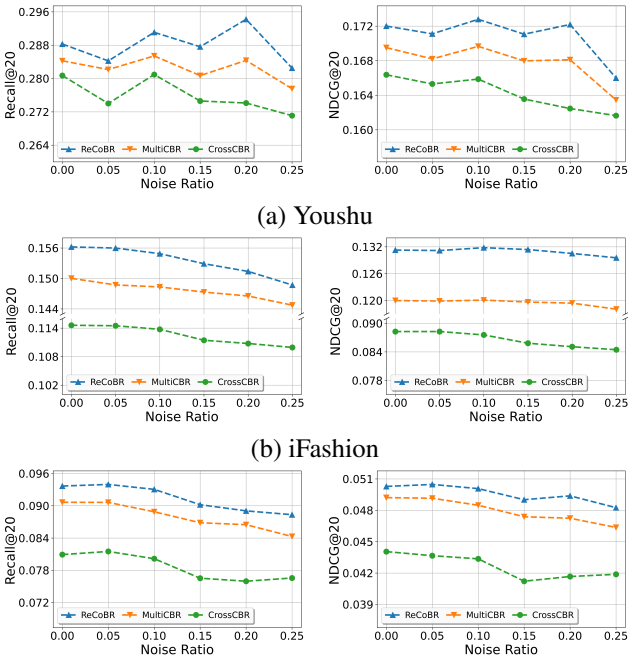
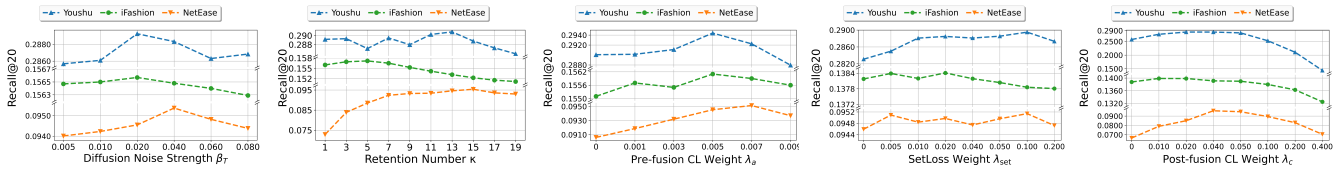


Figure 3: Recall@20 and NDCG@20 under injected U-B graph noise on Youshu, iFashion, and NetEase. We add random edges at ratios from 0% to 25% in 5% increments relative to each user's original training interactions, retaining all original edges.



(a) (b) (c) (d) (e)

Figure 4: Hyperparameter sensitivity in Recall@20 on Youshu, iFashion, and NetEase. (a) Diffusion noise variance endpoint β_T. (b) U-B retention limit κ. (c) Pre-fusion contrastive weight λ_a. (d) SetLoss weight λ_{set}. (e) Post-fusion contrastive weight λ_c. Broken y-axes use different scales for the datasets; candidate values are displayed at equally spaced horizontal positions. Panels (d) and (e) report one run per setting.

These results suggest that explicit cross-view learning is particularly useful when direct bundle feedback is limited, allowing U-I behavior and B-I composition to provide additional learning signals for bundle preference modeling.

Robustness to Noisy U-B Interactions. We compare ReCoBR with CrossCBR and MultiCBR using the same perturbed U-B input graph for each dataset and noise ratio. Ranking supervision, U-I and B-I relations, and evaluation data remain unchanged. We rerun ReCoBR's full training pipeline, including U-B pretraining. Figure 3 shows that ReCoBR maintains the highest Recall@20 and NDCG@20 across all datasets and tested noise ratios, showing that it retains its recommendation advantage when additional random connections interfere with U-B graph propagation.

4.5 Hyperparameter Sensitivity

Figure 4 reports ReCoBR's sensitivity to five key hyperparameters. We vary one parameter at a time while keeping the others fixed. For β_T, intermediate noise levels yield better performance than the extremes, suggesting that preference recovery benefits from moderate perturbation without excessively disrupting historical information. For κ, iFashion favors smaller retained neighborhoods, whereas NetEase benefits from retaining more connections. This difference highlights the need to balance neighborhood coverage with preference relevance when selecting propagation connections. Nonzero λ_{set} weights improve performance on Youshu, while iFashion and NetEase show only small, nonmonotonic changes. Increasing this weight does not consistently yield further gains. Moderate nonzero values of λ_a and λ_c improve performance, while larger weights can reduce it. The decline is especially pronounced for λ_c on Youshu and NetEase, suggesting the need to balance contrastive regularization with the ranking objective.

5 Conclusion

In this paper, we proposed ReCoBR to address noisy and sparse U-B feedback in bundle recommendation. ReCoBR refined historical bundle preferences through user-conditioned diffusion with a bundle-set consistency constraint and used the refined preferences to reconstruct the U-B propagation graph. The reconstructed U-B view then anchored pre-fusion contrastive learning, enabling auxiliary behavioral and compositional information to support bundle preference modeling. Experiments on three public datasets demonstrated ReCoBR's advantages over state-of-the-art baselines and the effectiveness of its key designs.

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